

Practical Generative AI for Beginners

Professional Training Curriculum & Learning Roadmap

Beginner-friendly | Instructor-led | Hands-on | Enterprise-aware

A structured 2-month program designed to take learners from essential development foundations to practical Generative AI applications using local LLMs, RAG and AI Agents - with DevOps, DevSecOps and AI Security awareness built into the journey.

Duration	2 Months
Schedule	2 Hours per Day
Learning Time	Approx. 60-70 Hours
Format	Concepts + Live Demos + Labs + Assignments + Capstone

Course Overview

This course focuses directly on Generative AI and does not require prior Machine Learning or Deep Learning knowledge. Learners build the programming, version-control, DevOps and security foundations needed to work confidently with modern LLM applications.

Who Should Attend

Students, freshers, career switchers, software developers, QA engineers, DevOps engineers, security professionals and technology enthusiasts who want a practical entry point into Generative AI.

Prerequisites

Basic computer knowledge is sufficient. No prior AI experience is required. Basic Python required for LLM application development is taught as part of the program.

Course Highlights

Local LLM hands-on practice with Ollama; Prompt Engineering; LangChain; ChromaDB and RAG; LangGraph and AI Agents; Git and GitHub; Linux and DevOps foundations; DevSecOps and AI Security awareness; guided labs; assignments; and an end-to-end capstone project.

Learning Roadmap

Stage	Learning Path	What the learner gains
1	Git & GitHub Fundamentals	Version control and collaboration
2	Basic Python for LLMs	Programming foundation for GenAI
3	Linux & DevOps Foundations	Environments, CLI and Docker awareness
4	DevSecOps Awareness	Secure development and AI security mindset
5	Generative AI Fundamentals	LLMs, tokens, context and use cases
6	Prompt Engineering + Ollama	Prompting and local LLM execution
7	LangChain	Reusable LLM application workflows
8	ChromaDB + RAG	Grounded, document-aware AI applications
9	LangGraph + AI Agents	Stateful and multi-step agent workflows
10	Capstone Project	Integrated end-to-end GenAI solution

Detailed Training Curriculum

Module 1 - Generative AI Fundamentals

Module focus: Build a clear foundation in Generative AI without requiring ML or Deep Learning prerequisites.

Topics covered

- What Generative AI is and where it is used
- Large Language Models (LLMs) and how users interact with them
- Tokens, context windows, inference and model limitations
- Hallucinations, grounding and responsible use
- Common enterprise and productivity use cases

Hands-on lab: Explore different LLM responses and identify strengths, limitations and suitable use cases.

Module 2 - Basic Python for LLMs

Module focus: Learn only the Python concepts needed to start building practical LLM applications.

Topics covered

- Variables, data types, conditions and loops
- Functions and reusable code
- Lists, dictionaries and JSON
- Reading and writing files
- Modules, packages, pip and virtual environments
- Calling APIs from Python at a beginner level

Hands-on lab: Create a small Python program that reads input, processes JSON and prepares data for an LLM workflow.

Module 3 - Git & GitHub Fundamentals

Module focus: Use professional version-control practices from the beginning of the GenAI journey.

Topics covered

- Repositories, clone, add, commit, push and pull
- Branches and basic merge workflow
- README and project structure
- GitHub collaboration and pull-request awareness
- Basic .gitignore and safe repository practices

Hands-on lab: Create a GitHub repository for the course project and manage changes through commits and branches.

Module 4 - Linux & DevOps Foundations

Module focus: Understand the development environment and operational concepts that support reliable AI applications.

Topics covered

- Linux terminal navigation and file operations

- Permissions and environment variables
- Package and process basics
- Development environments and dependency consistency
- Docker concepts and why containers matter
- Introduction to build, test and deployment flow

Hands-on lab: Practice Linux commands and understand how an AI application can be packaged consistently.

Module 5 - DevSecOps & AI Security Awareness

Module focus: Introduce security at the right level for beginners so secure habits become part of GenAI development.

Topics covered

- Secure coding mindset
- Secrets and API-key protection
- Dependency and open-source risk awareness
- Prompt injection and unsafe-input awareness
- Sensitive-data and privacy considerations
- Responsible AI and basic secure AI development practices

Hands-on lab: Review an example GenAI project for exposed secrets, risky dependencies and prompt-injection scenarios.

Module 6 - Prompt Engineering

Module focus: Learn how to communicate effectively with LLMs and produce more consistent, useful outputs.

Topics covered

- Prompt anatomy: role, task, context and constraints
- Zero-shot and few-shot prompting
- Role-based and instruction prompting
- Structured outputs and formatting
- Prompt templates and reusable prompts
- Basic response evaluation and iterative improvement

Hands-on lab: Create and compare prompts for summarization, extraction, classification and structured output.

Module 7 - Ollama & Local LLMs

Module focus: Run open-source LLMs locally so learners can experiment without depending on a hosted application framework.

Topics covered

- Ollama installation and setup
- Downloading and running local models
- Model selection basics
- CLI interaction and model parameters
- Ollama REST API

- Connecting Python applications to a local LLM

Hands-on lab: Install Ollama, run a local model and invoke it from Python.

Module 8 - LangChain Fundamentals

Module focus: Move from direct model calls to reusable application components and LLM workflows.

Topics covered

- LangChain purpose and architecture at a beginner level
- Prompt templates
- Model invocation and output handling
- Output parsers
- Simple chains and reusable workflows
- Integration with Ollama

Hands-on lab: Build a simple LangChain application backed by a local LLM.

Module 9 - ChromaDB & Retrieval-Augmented Generation (RAG)

Module focus: Build AI applications that answer from selected documents instead of relying only on model knowledge.

Topics covered

- Why RAG is needed
- Document loading and text chunking
- Embeddings introduced in the context of RAG
- Vector databases and ChromaDB
- Semantic retrieval
- Prompt + retrieved context flow
- Grounding and source-aware response concepts

Hands-on lab: Create a local document Q&A; / knowledge assistant using ChromaDB and Ollama.

Module 10 - LangGraph & AI Agents

Module focus: Introduce agentic application patterns after learners understand prompts, LLM applications and RAG.

Topics covered

- What an AI Agent is and when to use one
- Workflow vs agent concepts
- State and multi-step execution
- Nodes, edges and conditional flow in LangGraph
- Tool-use concepts
- Human-in-the-loop awareness
- Simple agent orchestration patterns

Hands-on lab: Build a small stateful LangGraph workflow that performs multiple steps to complete a task.

Module 11 - Capstone Project

Module focus: Combine the complete learning path into a portfolio-ready practical project.

Topics covered

- Project planning and Git repository structure
- Local LLM integration with Ollama
- Prompt templates and LangChain workflow
- Document retrieval using ChromaDB and RAG
- Optional LangGraph agent workflow
- Basic DevOps and DevSecOps practices
- Testing, documentation and demonstration

Hands-on lab: Build and demonstrate a secure local AI Assistant using the technologies covered in the course.

Hands-on Lab Journey

Lab	Theme	Practical outcome
Lab 1	Development setup	Install/configure Git, Python and VS Code; create the course repository.
Lab 2	Python for LLMs	Practice functions, JSON, file handling and virtual environments.
Lab 3	Git workflow	Commit, branch, merge and maintain a clean project repository.
Lab 4	Linux & DevOps basics	Use essential Linux commands and explore environment/container concepts.
Lab 5	Secure GenAI basics	Identify exposed secrets, dependency risks and prompt-injection examples.
Lab 6	Prompt Engineering	Design and compare zero-shot, few-shot, role and structured prompts.
Lab 7	Local LLM with Ollama	Install Ollama, run a model and call the local API from Python.
Lab 8	LangChain application	Create a reusable prompt-to-model workflow.
Lab 9	RAG with ChromaDB	Build a local knowledge assistant using learner-provided documents.
Lab 10	LangGraph Agent	Create a stateful multi-step workflow / simple agent.
Capstone	Secure Local AI Assistant	Integrate Git, Ollama, LangChain, ChromaDB/RAG and LangGraph with basic DevSecOps practice

Why DevOps & DevSecOps Matter for GenAI

DevOps gives learners the practical foundation to manage source code, environments, dependencies and deployment workflows. **DevSecOps** adds security practices such as protecting secrets, reviewing dependencies and considering security throughout development. For beginners, these are foundation and awareness topics - advanced CI/CD, Kubernetes and enterprise platform engineering are outside the core scope.

Learning Outcomes

- Understand Generative AI and LLM fundamentals without requiring ML or Deep Learning knowledge.
- Use basic Python, Git/GitHub and Linux skills in a GenAI project.
- Run and integrate local LLMs with Ollama.
- Create effective prompts and reusable LangChain workflows.
- Build a document-based RAG application with ChromaDB.
- Understand and build introductory LangGraph / AI Agent workflows.
- Apply foundational DevOps, DevSecOps and AI Security awareness.
- Complete an end-to-end capstone project suitable for demonstration.

About the Trainer

Mukul Malhotra

Senior Technology Leader, Application Security / DevSecOps practitioner and Corporate Trainer with 20+ years of technology experience. His training focus spans Application Security, DevSecOps, DevOps, Generative AI, AI Security, secure software development and practical enterprise implementation.

Training Approach

The program is designed around concept clarity followed by live demonstrations, guided hands-on labs, assignments and an integrated capstone project. The emphasis is on helping beginners understand not only how to use GenAI tools, but also how Git, DevOps, DevSecOps and security practices fit into real-world application development.

Core Training Areas

Generative AI | Local LLMs | Ollama | Prompt Engineering | LangChain | ChromaDB | RAG | LangGraph | AI Agents | Git & GitHub | DevOps | DevSecOps | Application Security | AI Security

Training Duration

2 Months | 2 Hours per Day | Approx. 60-70 Hours

Includes instructor-led learning, live demos, hands-on labs, assignments and capstone project activities.